

1. What is SQL?

Ans:- SQL (Structured Query Language) is a standard programming language used to store, retrieve, manage, and manipulate data in relational databases such as MySQL, PostgreSQL, Oracle, SQL Server, and SQLite.

2. What is difference between Data and Information?

Data	Information
Data is a collection of raw facts and figures.	Information is processed and organized data that has meaning.
It is unorganized and unprocessed.	It is organized, processed, and meaningful.
Data alone may not help in making decisions.	Information helps in making decisions.
It is the input for processing.	It is the output after processing data.
Example: 85, 90, 78	Example: "The student's average score is 84.3%."

Example

Suppose a student's marks are:

- Math: 85
- English: 90
- Science: 78

These individual marks are data because they are raw facts.

After calculating the average:

- Average = 84.3%
- Result = Passed with First Division

This processed result is information because it provides meaningful insight.

In one sentence

- Data = Raw facts and figures.
- Information = Processed data that is meaningful and useful for decision-making.

3. What is Database and DBMS?

Ans:- A **Database** is an organized collection of related data that is stored electronically so it can be easily accessed, managed, and updated.

Example:

A college database may store:

- Student details
- Teacher details
- Course information
- Marks and attendance

Instead of keeping records on paper, all this information is stored in a database.

DBMS (Database Management System) is software that allows users to create, store, retrieve, update, and manage databases.

It acts as an interface between the user and the database.

Examples of DBMS:

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite

4. What is Relational Database Management System?

Ans:- A Relational Database Management System (RDBMS) is a type of Database Management System (DBMS) that stores data in the form of tables (relations) consisting of rows and columns. It uses relationships between tables to organize and manage data efficiently.

Example:- Consider two tables:

Students

Student_ID	Name	Class
101	Riya	BCA
102	Aman	BCA

Marks

Student_ID	Subject	Marks
101	SQL	85
102	SQL	90

Here, Student_ID is used to relate the two tables. This relationship allows you to retrieve information such as a student's name along with their marks.

5. State at least 5 names of Database Management System or Software.

- MySQL
- Oracle Database
- Microsoft SQL Server
- PostgreSQL
- SQLite